

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Revision Test 2026

MATHEMATICS — SSC-I (9th Grade)

Syllabus: Unit 1 – Real Numbers, Unit 2 – Logarithms, Unit 4 – Factorization and Algebraic Manipulation, Unit 5 – Linear Equations and Inequalities, Unit 7 – Coordinate Geometry

Total Marks: 50

Time Allowed: 2 Hours 15 Minutes

Version 18

General Instructions:

1. Attempt **ALL** questions. **There are no choices in this paper.**
2. Section A carries 10 marks, Section B carries 24 marks and Section C carries 16 marks.
3. In Section A, write the **letter** of the correct option clearly. Do the rough working for each MCQ on your answer sheet. Cutting or overwriting an answer will not be credited.
4. Write all working for Section B and Section C. Marks are awarded for correct working, not only for the final answer.
5. Answer **every part** of each question. Where a question says *verify*, *show that*, *name* or *state*, write that step and the final conclusion in words.
6. Use of a calculator is not allowed.

SECTION A — Multiple Choice Questions (10 × 1 = 10 Marks)

Attempt **ALL** questions. Write the letter of the correct option.

#	Question	A	B	C	D	Answer
1	$\sqrt{98} + \sqrt{8}$ is equal to:	$\sqrt{106}$	$9\sqrt{2}$	$5\sqrt{2}$	$10\sqrt{2}$	(A) (B) (C) (D)
2	$\log 6 + \log 5$ is equal to:	$\log 30$	$\log 11$	$\log 1.2$	$\log 6 \times \log 5$	(A) (B) (C) (D)
3	$\log_3 (1/27)$ is equal to:	3	$1/3$	-3	$-1/3$	(A) (B) (C) (D)
4	The factors of $27m^3 + 64n^3$ are:	$(3m + 4n)(9m^2 + 12mn + 16n^2)$	$(3m - 4n)(9m^2 + 12mn + 16n^2)$	$(3m + 4n)^3$	$(3m + 4n)(9m^2 - 12mn + 16n^2)$	(A) (B) (C) (D)
5	The factors of $27 - 8y^3$ are:	$(3 - 2y)(9 + 6y + 4y^2)$	$(3 - 2y)(9 - 6y + 4y^2)$	$(3 + 2y)(9 - 6y + 4y^2)$	$(3 - 2y)^3$	(A) (B) (C) (D)
6	$ -9 - -4 $ is equal to:	-13	5	-5	13	(A) (B) (C) (D)
7	The solution of $-4 \leq 2x - 2 \leq 6$ is:	$1 \leq x \leq 4$	$-3 \leq x \leq 4$	$-1 \leq x \leq 4$	$-1 \leq x \leq 2$	(A) (B) (C) (D)
8	If $x - 3 < 5$, then $x < 8$. The property of inequality used is the:	additive property	multiplicative property	transitive property	trichotomy property	(A) (B) (C) (D)
9	If the ordered pairs $(2a - 1, b + 3)$ and $(9, 5)$ are equal, then the value of a is:	9	4	2	5	(A) (B) (C) (D)
10	The mid-point of the line segment joining $(-5, 4)$ and $(9, -8)$ is:	$(7, -6)$	$(2, -2)$	$(-2, 2)$	$(4, -4)$	(A) (B) (C) (D)

SECTION B — Short Questions (6 × 4 = 24 Marks)

Attempt **ALL** questions. There are no choices.

Part	Question	Marks
2(i)	(a) Simplify: $\sqrt{63} + \sqrt{28} - \sqrt{7}$ [2] (b) Evaluate: $(27)^{2/3} \times (9)^{-1/2}$ [2]	4
2(ii)	Given that $\log 2 = 0.3010$ and $\log 3 = 0.4771$: (a) Evaluate log 72 . Show every step and name the law of logarithms used at each step. [3] (b) Write the characteristic of $\log 0.0567$. [1]	4
2(iii)	Factorize: (a) $1 - 343z^3$ [2] (b) $x^3 - 6x^2y + 12xy^2 - 8y^3$ [2]	4
2(iv)	(a) Solve $ x - 3 + 4 = 9$ and verify both solutions. [2] (b) Solve $3(x - 1) \geq 2x + 4$ and state the property of inequality used in each step. [2]	4
2(v)	Using the distance formula, show that the points A(-2, -1), B(1, 3) and C(7, 11) are collinear. Write your conclusion.	4

2(vi)	Simplify: $1/(x - 2) - 4/(x^2 - 4)$	4
-------	-------------------------------------	---

SECTION C — Long Questions ($2 \times 8 = 16$ Marks)

Attempt **ALL** questions. There are no choices.

Part	Question	Marks
3(i)	(a) Find the square root of $x^4 - 8x^3 + 22x^2 - 24x + 9$ by the division method. [4] (b) Find the HCF and the LCM of $x^2 - 11x + 28$ and $x^2 - 3x - 28$ by factorization, and verify that $\text{HCF} \times \text{LCM} = \text{product of the two polynomials}$. [4]	8
3(ii)	Two houses stand at the points P(-4, 3) and Q(6, -7). A lamp post is fixed at the mid-point M of PQ so that both houses get equal light. (a) Find the coordinates of M. [2] (b) Show that $PM = QM$. Give the distance in simplest radical form. [2] (c) A third house stands at R(8, -1). Show that R is at the same distance from the lamp post as P and Q. [2] (d) Find PQ and show that $PM + MQ = PQ$. What does this tell you about the points P, M and Q? [2]	8

Invigilator's Signature: _____

Candidate's Signature: _____

Units covered: 1 Real Numbers • 2 Logarithms • 4 Factorization and Algebraic Manipulation • 5 Linear Equations and Inequalities • 7 Coordinate Geometry

— END OF QUESTION PAPER —